

# Ethan Weimer-Kopf

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## EDUCATION

### Northwestern University (NU) McCormick School of Engineering

Evanston, IL

Bachelor of Science in Mechanical Engineering

Expected Graduation June 2029

Cumulative GPA: 3.9/4.0; Dean's List 2025-2026

Relevant Coursework: Physics, Engineering Analysis (Statics and Dynamics), Materials Science Engineering, Chemistry, Linear Algebra

### Johns Hopkins University Whiting School of Engineering

Baltimore, MD

Explore Engineering Innovation Course | GPA: 4.0

## CERTIFICATIONS & TRAINING

ASME Y14.5 GD&T Fundamentals | SolidWorks Design Associate (CSWA) | OSHA 10 Hr. General Industry Certification

## TECHNICAL EXPERIENCE

### NU Formula Society of Automotive Engineers Team

*Suspension Sub-team – Project Lead*

Sep 2025 - Present

- Redesigning hubs into a more compact, lightweight, inboard corner architecture using FEA-optimized load paths; reduced unsprung mass by **~45%** while limiting the manufacturing cost increase to **~10%** and maintaining structural integrity across all load cases. Used GD&T and close collaboration with teammates to ensure seamless direct integration with 8+ suspension components.
- Designed and implemented a custom spring-damping system inside the rear half shafts (split axle drive train) to dissipate energy and mitigate axial play, reducing measured axial play by an estimated **~30%** and resulting in smoother cornering with reduced wear on drive train components.
- Manufactured **30+ components** in house using manual and CNC mill and lathe, modeled core suspension components in SolidWorks, and conducted FEA and tensile strength testing for carbon fiber and steel wishbones, cutting projected per-part costs by **~15%**. Documented savings in cost reports and employed GD&T to create detailed dimensioned drawings of core suspension components for FSAE competition.

### NCSU Dep. Mechanical and Aerospace Engineering – Yeh Research Group

Jul 2026 - Present

- Researched unsteady fluid dynamics: Utilized TecPlot360 to generate visualizations from **2TB** of asymmetrical and unsteady airflow data over a delta-wing; also conducting CFD simulations on frisbee golf disc trajectories to study evolution of unsteady airflow during flight.

### NU Space Technology and Rocketry Society

Sep 2025 - Present

- Heat transfer lead on a 5-person power generation sub-team: designed thermal systems, including a waste-heat recovery concept projected to capture **~25%** of otherwise-lost energy from NASA's Fission Surface Power system, for a NASA RASC-AL proposal supporting a long-term, self-sustaining lunar habitat. Led the investigation of waste heat recovery, developing concepts for materials selection, heat collection, and large-scale thermal distribution designed to operate across a **200°C+** lunar day-night thermal swing.

## MECHANICAL DESIGN PROJECTS

### Zero Torque Putter

Feb 2026 – Present

- Designed and prototyped a modular zero-torque golf putter in SolidWorks, using mass properties analysis to optimize center of mass and weight distribution while aligning the shaft axis with the center of mass to eliminate torque during the putting stroke, improving putting accuracy by an estimated **~40%** in testing. Conducted Motion Analysis Simulation across **5+ design iterations** and validated results against real-world testing.
- Fabricated and iterated **3 prototypes** using 3D printed components and machined steel weights, achieving a functional design with interchangeable shaft inserts that maintain shaft-axis alignment through the center of mass across a range of shaft lengths.

### Wheelchair-Accessible Ball Launcher

Apr 2026 – June 2026

- Designed and implemented a custom ball-launching mechanism to help a local child with spinal muscular atrophy play catch with his classmates. Modeled full prototypes in SolidWorks before 3D printing components and running **100+ test trials**, increasing launch distance by an estimated **~40%** over the initial design while minimizing safety hazards.

### Modular Vendor Kiosk

Jan 2026 – Apr 2026

- Developed a portable, modular kiosk for pop-up market vendors in the local area. Interviewed the client and 6 local vendors, synthesizing feedback into design requirements and a target unit cost **under \$175**. Drew sketches and built mockups to communicate ideas to teammates and clients; presented the prototype and **90-page** related research paper to 400+ attendees at a poster fair.

## ADDITIONAL EXPERIENCE

### NU Li-Ion Coin Cell Project

Sep 2025 – Dec 2025

- Co-led a project on Li-Ion coin cell production. Utilized electrochemical impedance spectroscopy and cycling tests to investigate the efficacy of varying separator thickness and electrolyte composition on reducing dendrite formation (which causes short-circuiting) in Li-Ion coin cells, achieving a **~60% decrease** in dendrite formation while incurring a deficit of only **~15%** ionic conductivity.

### Intern - NC State University REU Computational Data Science in Astrophysics

May 2024 - June 2024

- Utilized Computational Fluid Dynamics (CFD) and coded in Linux and Fortran under the guidance of Dr. John Blondin to process **~1.5 TB** of simulation data and create a 3D hydrodynamics model of the X-Ray Binary Star System Cyg X-1, refining understanding of black hole accretion dynamics.

### Mathnasium – Math Instructor

Summer 2026 | Jul 2024 – Sep 2025

- Tutored **~50 students** in material from counting to calculus; conducted private sessions teaching topics such as Calculus II and SAT Prep.

### USA Invitational Young Physicists' Tournament (USAYPT)

Aug 2023 – May 2025

*Project Leader | 2025 Tournament Champion | 2024 Tournament Finalist*

- Led a **7-person team** in year-long research on advanced physics problems involving theory, experimentation, modeling, and prototyping.
- 2025: Modeled the aerodynamics of disc golf discs, analyzing how design characteristics influence aerodynamic forces and flight trajectories.
- 2024: Created a mathematical model to reconstruct a 3D map of the Moon's surface using images captured with a standard smartphone camera.

## LEADERSHIP & ACTIVITIES

Golf Team Captain (3 yrs.) | Physics Team Project Lead (2 yrs.) | Ultimate Frisbee Team Captain/Middle School Coach (2 yrs.)

## SKILLS

**CAD & Analysis:** SolidWorks, FEA, Motion Analysis, GD&T (ASME Y14.5), DFM | **Documentation:** Technical Writing, LaTeX

**Programming & Computation:** MATLAB, Python, Java, Fortran, Linux

**Manufacturing:** Manual & CNC Milling, CAM, 3D Printing | **Experimental:** SEM, XRD, Potentiostat, Li-Ion Coin Cell Assembly